



Regression Mini Project Guidelines

Prepared by :

Aryam Alanazi
Deem Ali Alnughmush
Fay Alsalhi
Rahaf Saud Alharbi
Renad Albehadal

Presented to :

Dr.Renad Alsweed





Introduction :

This project applies straightforward machine learning techniques to analyze and predict coffee quality based on a range of physical and chemical characteristics. The data collection involves multiple attributes that characterize coffee origin, processing type, and cup quality ratings obtained from different regions. The main goal is to develop a model for regression that can predict the coffee quality score from its measurable attributes. For this purpose, the project follows the standard machine learning pipeline:

data collection and preparation, exploratory data analysis, model selection and training, evaluation, and final interpretation of results.

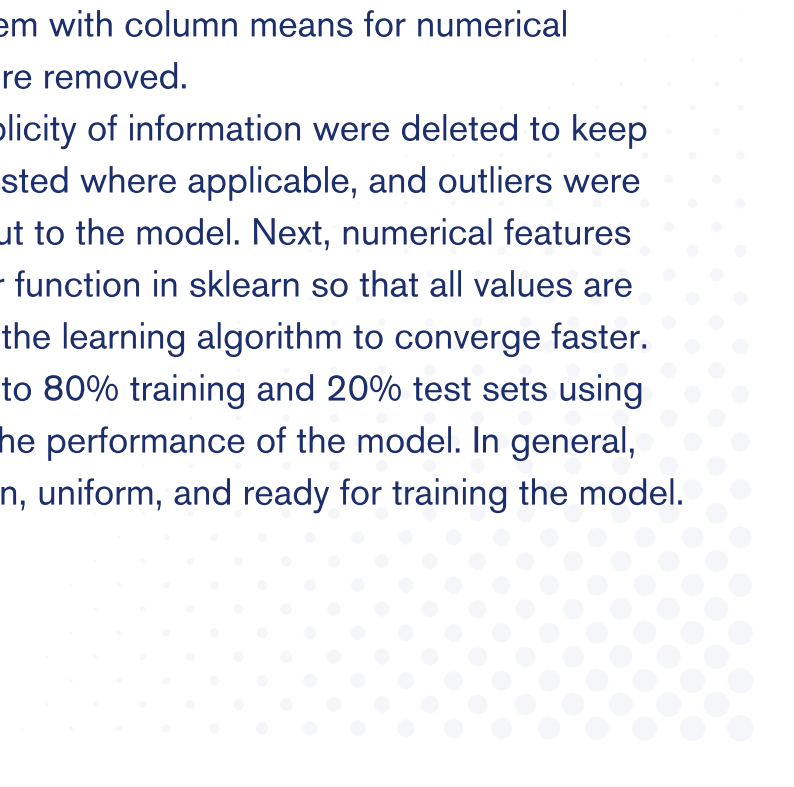
In this paper, we would like to demonstrate how regression techniques and data preprocessing can be applied to a real-world dataset in order to derive conclusions about determinants of coffee quality.

Data Preprocessing :

In this step, the merged coffee quality dataset was imported and analyzed using Pandas. Initial analysis showed a series of numeric and categorical features related to coffee origin, processing, and quality scores.

Missing values were handled by imputing them with column means for numerical features, and categorical inconsistencies were removed.

Irrelevant columns such as identifiers or duplicity of information were deleted to keep the dataset clean. Data types were also adjusted where applicable, and outliers were recognized and resolved to feed cleaner input to the model. Next, numerical features were standardized using the StandardScaler function in sklearn so that all values are brought to an equivalent range, which helps the learning algorithm to converge faster. Finally, the structured dataset was divided into 80% training and 20% test sets using train_test_split for unbiased assessment of the performance of the model. In general, this phase guaranteed the data set was clean, uniform, and ready for training the model.



Model Selection and Training :

In this stage, three regression models were selected and trained to predict the Total Cup Points score based on the coffee dataset.

The models used were Linear Regression, Ridge Regression, and Random Forest Regressor.

- Linear Regression was chosen as a simple baseline model to learn the linear relationship between features and the target variable.
- Ridge Regression adds regularization to the linear model to prevent overfitting and improve generalization.
- Random Forest Regressor was used as a non-linear ensemble model to compare the performance against the linear approaches.

Each model was trained using the training subset (X_{train} , y_{train}) by applying the `fit()` function. After training, all models were ready for evaluation.

Evaluation Results :

After training the three regression models, their performance was tested using a separate test dataset that was not used during training. The evaluation was based on three common metrics: Mean Absolute Error (MAE), Mean Squared Error (MSE), and R-squared (R^2). The results were as follows:

Linear Regression: MAE = 0.0097, MSE = 0.0011, R^2 = 1.0000

Ridge Regression: MAE = 0.0095, MSE = 0.0010, R^2 = 1.0000

Random Forest Regressor: MAE = 0.5779, MSE = 16.1273, R^2 = 0.4936

Both Linear Regression and Ridge Regression achieved very high accuracy with very small errors. This means that the connection between the coffee features and the total quality score is mostly linear. The Ridge model gave slightly better results because it includes a small adjustment that prevents the model from overfitting and makes it more stable. On the other hand, the Random Forest Regressor did not perform as well as the linear models, which suggests that this dataset works better with simpler linear approaches.

The scatter plots also showed that the predictions from the two linear models were very close to the real values, while the Random Forest predictions were more spread out around the reference line.

The following figures show the model predictions and performance comparison. Linear and Ridge models clearly performed better than Random Forest.

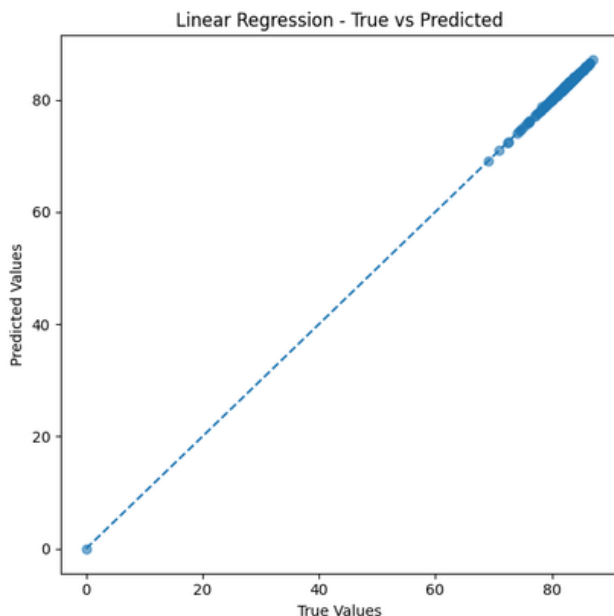


Figure 1: True vs Predicted Values for Linear Regression

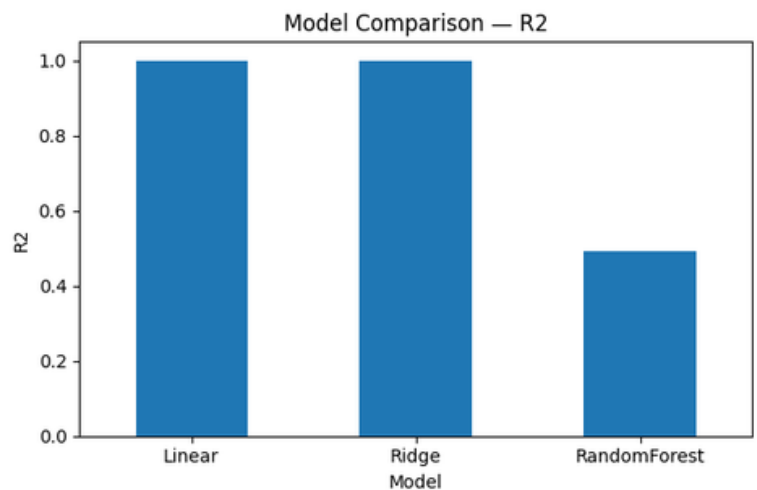


Figure 2: Comparison of R² Scores for All Models

Discussion and Findings:

The analysis confirms that careful preprocessing and choosing the right model type are key to achieving accurate results. By handling missing values, scaling numerical data, and removing irrelevant columns, the dataset became well-structured and consistent, allowing the algorithms to learn effectively.

The findings highlight that coffee quality can be predicted accurately using simple linear models, as most of the quality indicators like Aroma, Flavor, Acidity, and Body have direct and consistent influence on the total cup score.

Ridge Regression's slightly improved stability supports the importance of controlling correlated features. Although Random Forest Regressor did not outperform the linear models, its inclusion was valuable for comparison and helped confirm that the coffee quality prediction problem does not require complex non-linear methods.

Overall, this experiment demonstrates how a clean dataset and well-selected regression models can achieve reliable and interpretable predictions in a real-world scenario.